

10. Illustrate the concept of inter-process communication using message queue with a C program.

Code:

```
main.c
1 // message_queue.c
2 #include <stdio.h>
3 #include <sys/ipc.h>
4 #include <sys/msg.h>
5 #include <string.h>
6
7 struct msgbuf {
8     long mtype;
9     char mtext[100];
10 };
11
12 int main() {
13     key_t key = 1234;
14     int msgid;
15     struct msgbuf message;
16
17     msgid = msgget(key, 0666 | IPC_CREAT);
18
19     message.mtype = 1;
20
21     printf("Enter message: ");
22     fgets(message.mtext, sizeof(message.mtext), stdin);
23
24     msgsnd(msgid, &message, sizeof(message.mtext), 0);
25
26     msgrcv(msgid, &message, sizeof(message.mtext), 1, 0);
27
28     printf("Received Message: %s", message.mtext);
29
30     msgctl(msgid, IPC_RMID, NULL);
31
32     return 0;
33 }
```

Output:

```
Output
Enter message: Operating Systems
Received Message: Operating Systems

=== Code Execution Successful ===
```

